Global HIV Treatment Progress: the 95-95-95 Targets

An end-to-end analytics project that cleans UNAIDS estimates in **Power Query** and presents progress toward the 95-95-95 targets in a four-page **Power BI** dashboard, with measures written in **DAX**.



Business question

How close is the world to the 95-95-95 targets, where are the gaps, and who is being left behind?

The targets state that, by 2025:

1. **95%** of people living with HIV know their status (1st 95)
2. **95%** of those who know their status are on treatment (2nd 95)
3. **95%** of those on treatment have a suppressed viral load (3rd 95)

Together these mean about 86% of all people living with HIV virally suppressed (95% x 95% x 95%).

Key findings

> Numbers are Estimates from UNAIDS 2026 data, Global, all ages, unless stated.

1. **Global progress (2025):** an estimated **88%** know their status, **89%** of those are on treatment, and **95%** of those on treatment are virally suppressed. The 3rd 95 has reached its target; the 1st 95 is the weakest step, about **7 points** short.
2. **Progress over time:** the 1st 95 rose from **69% (2015)** to **88% (2025)**; the 2nd 95 from **68% (2015)** to **89% (2025)**. The 95-95-95 series are only reported from 2015.
3. **Regional gaps:** [Region with the largest gap] is furthest from target on [step], at [X]%. [Region(s)] have reached 95 on [step].
4. **Who is left behind:** [Group, for example children] trails adults on [step] by about [X] points. [Women vs men observation].
5. **Uncertainty matters:** the global 3rd 95 estimate is 95%, but its Low to High range is wide (about **72% to 98%**), so reaching the target is less certain than the headline suggests.

6. **Pace of progress:** year-on-year gains [slowed / held steady] after 2020 for [step].

Recommendations

- Prioritize **testing and diagnosis** where the 1st 95 gap is largest, because people who do not know their status cannot start treatment.
- Direct support first to [regions] and [groups] with the largest gaps shown on pages 2 and 3.
- Report estimates **with their ranges**, so a result at target is not read as certain when the uncertainty is wide.

Dashboard pages

Page	Question it answers	Main visuals
1. Global progress	How far has the world come on each of the three steps?	KPI cards (latest estimate and gap to 95), trend line with the 95 target, takeaway text
2. Regional gaps	Which regions and countries are closest to or furthest from 95?	Regional bar chart, red-amber-green heat table, country map
3. Who is left behind	Which groups and which step lag, and how certain are the estimates?	Group comparison (children, women, men), bottleneck chart (gap to 95), estimate with Low to High range, year-on-year change
4. Data notes	How should the numbers be read?	Source, definitions, cleaning rules and limitations


Page 2: Regional gaps

Page 3: Who is left behind

Data

- **Source:** UNAIDS 2026 estimates, *HIV estimates with uncertainty bounds, 1990 to 2025* ([add link to the UNAIDS data page]).
- **Sheet used:** `HIV-Test-&-Treat_ByArea` (Global, regions and countries, 2010 to 2025).
- **Raw shape:** wide, with 3 title rows, 3 header rows built from merged cells, about 100 columns and about 2,944 data rows.
- **Clean shape:** one long table, `HIV_Data`, with **141,439 rows and 12 columns**.

Tools and skills

Tool	Used for
Excel	Inspecting the source and checking numbers against the dashboard
Power Query (M)	Rebuilding headers, unpivoting, splitting, type conversion, flags
Power BI	Data model, visuals, conditional formatting, maps
DAX	Measures for latest estimate, gap to target, range and year-on-year change

Data cleaning (Power Query)

The full step-by-step record is in `docs/data_cleaning_report.pdf`. In short:

1. **Kept the raw sheet untouched** as a reference query, and built a small `Headers` query from the three header rows (Keep top rows, Transpose, Fill Down, Merge with a `|` separator).
2. **Renamed about 100 columns in one step** using the built headers, instead of typing them by hand.
3. **Unpivoted** the 97 measurement columns into long format (one row per measurement), then **split** each name into `Indicator`, `Group` and `Statistic`.
4. **Converted text to numbers:** `...` became blank (missing is not zero), `>98` became 98 with a `>` flag, `1.3 m` became 1,300,000, and `680 000` became 680000.
5. **Added helper columns:** `Qualifier` (Exact, `>`, No data), `Unit` (Percent or Count), `Indicator_Short` and `Area_Level` (Global, Region, Country).
6. **Standardized labels** such as `(All ages)` and `All ages`.
7. **Validated** the result: row counts, spot checks of Global 2025 against Excel, duplicate and range checks.

Cleaning decisions

Raw value	Treatment	Why
`...`	Blank (null)	No data is not the same as zero
`>98`	98, flagged `>`	It is a lower bound, not an exact value
`1.3 m`	1,300,000	`m` means millions (rounded, so approximate)

Raw value	Treatment	Why
`680 000`	680000	The space is a thousands separator
Global, regions, countries	Separate `Area_Level`	Adding them together double counts

Key DAX measures

```
Estimate % = CALCULATE(AVERAGE(HIV_Data[Value_Num]), HIV_Data[Statistic] =
"Estimate", HIV_Data[Unit] = "Percent")

Latest Estimate % = CALCULATE([Estimate %], LASTNONBLANK(HIV_Data[Year],
[Estimate %]))

Gap to 95 = 95 - [Latest Estimate %]

YoY change = VAR CurrentYear = MAX(HIV_Data[Year]) VAR Prior =
CALCULATE([Estimate %], HIV_Data[Year] = CurrentYear - 1) RETURN
IF(ISBLANK(Prior), BLANK(), [Estimate %] - Prior)
```

Every visual is filtered to one `Area_Level`, one `Group` (usually All ages) and the step shown, so that numbers are never averaged across different things.

Challenges and what I learned

- **Flipping the whole table** created about 2,947 columns. I learned to fix structure on the small piece that needs it, in a separate query.
- **A dashboard card said 88 while a chart said 78.** Comparing the visual with the source showed that page filters had been removed, so the chart was averaging all areas and groups. I now check every number against the source.
- **A measure showed 202.50K%.** A multi-line paste had lost the first line, so the measure returned the year. I learned that a measure is named by what comes before its first equals sign.
- **Reading errors:** "Token Eof expected" meant a stray character after a finished formula, and "Token Literal expected" meant a formula started with something invalid.

Limitations

- Renaming columns is position-based, so reordered or added source columns could mislabel data after a refresh. Check a few columns after any update.
- Values capped at `>98` hide the true value.
- Counts shown in millions are rounded, so they are approximate.
- Some estimates have wide ranges, so an estimate at target can still be uncertain.
- The 95-95-95 series start in 2015 because earlier years were not reported.
- Averages across countries are unweighted, so regional comparisons use the published regional rows.

Repository structure

```
hiv-treatment-progress/
├── README.md
├── hiv_95_dashboard.pbix        # Power BI report
├── images/                      # Dashboard screenshots
│   ├── 01_global_progress.png
│   ├── 02_regional_gaps.png
│   ├── 03_who_is_left_behind.png
│   └── 04_data_notes.png
└── docs/
    └── data_cleaning_report.pdf # Step-by-step cleaning record
```

How to open the report

1. Download `hiv_95_dashboard.pbix` and open it in **Power BI Desktop** (free).
2. If the data source path has changed, go to **Transform data > Data source settings** and point it to your copy of the UNAIDS workbook.
3. Click **Refresh**, then check the Applied Steps in Power Query for errors.

Next steps

- **Project 3:** join this treatment data to AIDS mortality estimates, to test whether falling deaths follow rising treatment coverage.
- Add an overall viral-suppression KPI (2025 vs 2010) with the roughly 86% overall target.

About

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- Related project: E-commerce Customer Churn and Retention Analysis

Data: UNAIDS 2026 estimates. This project is for learning and portfolio purposes.